bi_suite_8datasets: counting correct guesses

What this folder shows

Every BI value starts from one number per model: how many of the attack traces it guessed correctly. This folder produces that number for the eight sets of traces in Table 7.

- `evaluate_tches20_pretrained_fixed_split.py`: runs all 80 trained TCHES20 models of one trace set and counts their correct guesses. This gives the six TCHES20 rows of Table 7.
- `evaluate_estranet_bi_suite.py`: does the same for a folder of saved Transformer models. This gives the ASCAD random-key and CHES-CTF-2025 rows. `paper_results/evaluate_estranet_suites.py` calls it with the settings used in the paper.
- `summarize_*.py`: turn the counts into table rows.
- `estranet_support/`: the loader for the CHES-CTF-2025 traces, the small patch used for training, and the training settings of every saved model.

A guess is correct when the model's top choice equals the true value of the key byte. Only this count feeds into BI.

Before you start

The Docker image (or the Python environment with `requirements-ml.txt`) and the TCHES20 download (`paper_results/fetch_tches20.sh`). For the Transformer part, also the source code, the models and the traces from guide 01, Level C.

Run: one set of TCHES20 models

Certificates are always computed on the attack traces, so pass `--split attack`:

```
T=datasets/raw/tches20/TCHES20V3_CNN_SCA
python -m bi_suite_8datasets.evaluate_tches20_pretrained_fixed_split \
  --dataset ascad_desync_0 --split attack \
  --models-dir $T/models/pretrained_models/models --dataset-dir $T/datasets --tches20-src-dir $T/src \
  --output-dir results/bi_suite_8datasets/tches20_fixed_split
```

```
$ T=datasets/raw/tches20/TCHES20V3_CNN_SCA; python -m bi_suite_8datasets.evaluate_tches20_pretrained_fixed_s
    plit --dataset ascad_desync_0 --split attack --models-dir $T/models/pretrained_models/models --dataset-d
    ir $T/datasets --tches20-src-dir $T/src --output-dir results/bi_suite_8datasets/tches20_fixed_split
[... earlier output omitted; key lines kept ...]
[...]
[54/80] zaid_ascad_desync_0_feature_scaling_m1_1_3.hdf5
[55/80] zaid_ascad_desync_0_feature_scaling_m1_1_4.hdf5
[56/80] zaid_ascad_desync_0_feature_scaling_m1_1_5.hdf5
[57/80] zaid_ascad_desync_0_feature_scaling_m1_1_6.hdf5
[58/80] zaid_ascad_desync_0_feature_scaling_m1_1_7.hdf5
[59/80] zaid_ascad_desync_0_feature_scaling_m1_1_8.hdf5
[60/80] zaid_ascad_desync_0_feature_scaling_m1_1_9.hdf5
[61/80] zaid_ascad_desync_0_feature_standardization_0.hdf5
[62/80] zaid_ascad_desync_0_feature_standardization_1.hdf5
[63/80] zaid_ascad_desync_0_feature_standardization_2.hdf5
[64/80] zaid_ascad_desync_0_feature_standardization_3.hdf5
```

```
[65/80] zaid_ascad_desync_0_feature_standardization_4.hdf5
[66/80] zaid_ascad_desync_0_feature_standardization_5.hdf5
[67/80] zaid_ascad_desync_0_feature_standardization_6.hdf5
[68/80] zaid_ascad_desync_0_feature_standardization_7.hdf5
[69/80] zaid_ascad_desync_0_feature_standardization_8.hdf5
[70/80] zaid_ascad_desync_0_feature_standardization_9.hdf5
[71/80] zaid_ascad_desync_0_no_preprocessing_0.hdf5
[72/80] zaid_ascad_desync_0_no_preprocessing_1.hdf5
[73/80] zaid_ascad_desync_0_no_preprocessing_2.hdf5
[74/80] zaid_ascad_desync_0_no_preprocessing_3.hdf5
[75/80] zaid_ascad_desync_0_no_preprocessing_4.hdf5
[76/80] zaid_ascad_desync_0_no_preprocessing_5.hdf5
[77/80] zaid_ascad_desync_0_no_preprocessing_6.hdf5
[78/80] zaid_ascad_desync_0_no_preprocessing_7.hdf5
[79/80] zaid_ascad_desync_0_no_preprocessing_8.hdf5
[80/80] zaid_ascad_desync_0_no_preprocessing_9.hdf5
{'dataset': 'ascad_desync_0', 'split': 'attack', 'scope_id': 'ascad_desync_0_tches20_m80_attack_fixed_split'
    , 'M': 80, 'n_te': 10000, 'p_obs': 0.0063, 'p_minus': 0.0025954814754893776, 'p_plus': 0.012 …

100%|██████████| 10000/10000 [00:01<00:00, 5911.31it/s]
[exit code 0, 20 s]
```

```
$ head -5 results/bi_suite_8datasets/tches20_fixed_split/ascad_desync_0_attack_success.csv && \
    cat results/bi_suite_8datasets/tches20_fixed_split/ascad_desync_0_attack_summary.csv
dataset,split,model_id,model_path,n_te,successes,success_rate,status,error,scope_id
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_0,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_0.hdf5,10000,42,0.0 …
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_1,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_1.hdf5,10000,38,0.0 …
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_2,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_2.hdf5,10000,39,0.0 …
ascad_desync_0,attack,noConv1_ascad_desync_0_feature_scaling_0_1_3,datasets/raw/tches20/TCHES20V3_CNN_SCA/mo
    dels/pretrained_models/models/noConv1_ascad_desync_0_feature_scaling_0_1_3.hdf5,10000,41,0.0 …
dataset,split,scope_id,M,n_te,p_obs,p_minus,p_plus,BI_minus,BI_obs,BI_plus,slack,R_margin,best_model,status,
    source_artifact
ascad_desync_0,attack,ascad_desync_0_tches20_m80_attack_fixed_split,80,10000,0.0063,0.0025954814754893776,0.
    012472640729235318,0.0,0.689567543950467,1.6749107576413271,0.006172640729235318,2.578648868 …
[exit code 0, 1 s]
```

For AES-RD add `--max-eval-traces 12500`, because the paper uses only the first 12,500 attack traces of that set.

Run: a set of Transformer models

The Transformer script takes the model settings on the command line:

```
python -m bi_suite_8datasets.evaluate_estranet_bi_suite --help
```

```
$ python -m bi_suite_8datasets.evaluate_estranet_bi_suite --help
usage: evaluate_estranet_bi_suite.py [-h] --project-root PROJECT_ROOT
                                     --dataset {ASCAD,CHES25} --dataset-name
                                     DATASET_NAME --scope-id SCOPE_ID
                                     --data-path DATA_PATH --checkpoint-dir
                                     CHECKPOINT_DIR
                                     [--checkpoint-indices CHECKPOINT_INDICES]
                                     [--skip-missing-checkpoints] --output-csv
                                     OUTPUT_CSV [--source-note SOURCE_NOTE]
                                     --input-length INPUT_LENGTH
                                     [--data-desync DATA_DESYNC]
                                     [--eval-batch-size EVAL_BATCH_SIZE]
                                     [--learning-rate LEARNING_RATE]
[... 28 lines omitted ...]
  --eval-batch-size EVAL_BATCH_SIZE
  --learning-rate LEARNING_RATE
  --n-layer N_LAYER
  --d-model D_MODEL
  --d-head D_HEAD
```

```
  --n-head N_HEAD
  --d-inner D_INNER
  --n-head-softmax N_HEAD_SOFTMAX
  --d-head-softmax D_HEAD_SOFTMAX
  --dropout DROPOUT
  --conv-kernel-size CONV_KERNEL_SIZE
  --n-conv-layer N_CONV_LAYER
  --pool-size POOL_SIZE
  --d-kernel-map D_KERNEL_MAP
  --beta-hat-2 BETA_HAT_2
  --model-normalization MODEL_NORMALIZATION
  --head-initialization HEAD_INITIALIZATION
  --softmax-attn, --no-softmax-attn
[exit code 0, 3 s]
```

The wrapper below fills in the paper's settings. It tests the eleven saved CHES-CTF-2025 models on 100,000 traces:

```
python -m paper_results.evaluate_estranet_suites --suites ches_ctf_2025 --skip-selected \
  --out-dir results/bi_suite_8datasets/estranet
```

```
$ python -m paper_results.evaluate_estranet_suites --suites ches_ctf_2025 --skip-selected --out-dir results/
    bi_suite_8datasets/estranet
[... earlier output omitted; key lines kept ...]
$ /usr/local/bin/python -m bi_suite_8datasets.evaluate_estranet_bi_suite --project-root external/EstraNet --
    dataset CHES25 --dataset-name ches_ctf_2025 --scope-id ches_ctf_2025_estranet_w3000_checkpoi …
ches_ctf_2025_estranet_ckpt1: successes=378 n_te=100000 success_rate=0.00378000 checkpoint=checkpoints/estra
    net/ches25_lx_w3000/trans_long-1
ches_ctf_2025_estranet_ckpt2: successes=382 n_te=100000 success_rate=0.00382000 checkpoint=checkpoints/estra
    net/ches25_lx_w3000/trans_long-2
ches_ctf_2025_estranet_ckpt3: successes=376 n_te=100000 success_rate=0.00376000 checkpoint=checkpoints/estra
    net/ches25_lx_w3000/trans_long-3
ches_ctf_2025_estranet_ckpt4: successes=402 n_te=100000 success_rate=0.00402000 checkpoint=checkpoints/estra
    net/ches25_lx_w3000/trans_long-4
ches_ctf_2025_estranet_ckpt5: successes=429 n_te=100000 success_rate=0.00429000 checkpoint=checkpoints/estra
    net/ches25_lx_w3000/trans_long-5
ches_ctf_2025_estranet_ckpt10: successes=388 n_te=100000 success_rate=0.00388000 checkpoint=checkpoints/estr
    anet/ches25_lx_w3000/trans_long-10
[...]

6250/6250 [==============================] - ETA: 0s
6250/6250 [==============================] - 107s 17ms/step
ches_ctf_2025_estranet_ckpt20: successes=406 n_te=100000 success_rate=0.00406000 checkpoint=checkpoints/estr
    anet/ches25_lx_w3000/trans_long-20

6250/6250 [==============================] - ETA: 0s
6250/6250 [==============================] - 106s 17ms/step
ches_ctf_2025_estranet_ckpt50: successes=423 n_te=100000 success_rate=0.00423000 checkpoint=checkpoints/estr
    anet/ches25_lx_w3000/trans_long-50

6250/6250 [==============================] - 106s 17ms/step
ches_ctf_2025_estranet_ckpt100: successes=415 n_te=100000 success_rate=0.00415000 checkpoint=checkpoints/est
    ranet/ches25_lx_w3000/trans_long-100

6250/6250 [==============================] - 111s 18ms/step
ches_ctf_2025_estranet_ckpt150: successes=406 n_te=100000 success_rate=0.00406000 checkpoint=checkpoints/est
    ranet/ches25_lx_w3000/trans_long-150

6250/6250 [==============================] - ETA: 0s
6250/6250 [==============================] - 107s 17ms/step
ches_ctf_2025_estranet_ckpt201: successes=413 n_te=100000 success_rate=0.00413000 checkpoint=checkpoints/est
    ranet/ches25_lx_w3000/trans_long-201
Wrote results/bi_suite_8datasets/estranet/estranet/ches_ctf_2025_estranet_w3000_success.csv
Wrote results/bi_suite_8datasets/estranet/estranet/ches_ctf_2025_estranet_w3000_success.summary.json
external/EstraNet/transformer.py:244: SyntaxWarning: "is" with a literal. Did you mean "=="?
  ks = 11 if l is 0 else self.conv_kernel_size
/usr/local/lib/python3.11/site-packages/keras/src/initializers/initializers.py:120: UserWarning: The initial
    izer RandomUniform is unseeded and being called multiple times, which will return identical  …
  warnings.warn(
```

```
[exit code 0, 20 min 40 s]
```

What to look for

- `<dataset>_attack_success.csv`: one row per model with its number of correct guesses. The largest number is what BI is computed from.
- `<dataset>_attack_summary.csv`: the BI of the whole set of models: the best success rate, its guaranteed upper limit, and both in bits.
- The Transformer script prints `successes=… n_te=100000` for each model. Models that guess about as well as chance (around 390 out of 100,000) can differ by one trace from machine to machine. The best model does not.
- `--split profiling` tests the models on the traces they were trained on. That is only a sanity check, not a certificate, and the script says so.